ENG-0021: System Postmortem for ledger-worker

Impact

The ledger-worker system, part of Project Fathom, experienced two recent hardware failure incidents, INC-004 and INC-041, both classified as Severity 2. Each incident resulted in an approximate resolution time of 19.5 hours. The hardware failures affected system availability and operational continuity, necessitating prompt response and recovery efforts. Given the system's medium criticality, these outages posed moderate risk to ongoing project activities and data integrity.

Timeline

- **INC-004** was reported by E-053, with the incident occurring prior to the resolution, which was completed after approximately 19.5 hours.
- **INC-041** was reported by E-054, with a similar resolution timeframe of 19.5 hours.
- Both incidents were addressed by the on-call engineer, Alyssa Long, Product Manager, based in Perth.
- The incidents occurred within the recent operational period, with the last resolution completed on or before the current document date, March 22, 2025.

Root Cause

Analysis indicates that both incidents stemmed from hardware failure within the ledger-worker system. The root cause category identified is hardware_failure. The hardware components involved are not specified, but the recurring nature suggests potential underlying hardware reliability issues or environmental factors impacting hardware stability.

Action Items

- Conduct a comprehensive hardware audit of ledger-worker components to identify potential weaknesses.
- Review environmental controls in the Perth location to mitigate hardware failure risks.
- Implement proactive monitoring and alerting for hardware health metrics.
- Schedule preventive maintenance based on hardware lifecycle and performance data.
- Document lessons learned and update incident response procedures to reduce resolution time for future hardware failures.

Additional Notes

The ledger-worker system is owned by Project Fathom, which was completed on March 13, 2025. The project is managed by Jamie Smith, DevOps Engineer, and overseen by the project owner department D-10. Alyssa Long, the current on-call engineer, has been involved in incident resolution, ensuring system stability. Continuous improvement efforts are recommended to enhance hardware resilience and minimize operational disruptions.

This document is intended for internal review and future reference to improve system reliability and incident management processes.